Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: GroundTruth
b. Team leader name: Lokesh Kadyan
c. Problem Statement: Sustainability & Social Impact: crop-loss insurance claims for smallholder farmers settled in minutes, not months, with satellite-verified evidence

Brief about the idea

“Our efforts were wiped out in three days.” Jaya Krishna Dharal, paddy farmer, Jumla, Nepal, after unseasonal rain flattened the harvest (Kathmandu Post, Nov 2025).

This is the season farmers are living right now: the Bhotekoshi floods washed away 1,806 hectares of farmland in Nuwakot, Nepal (Sep 2026). Three typhoons hit 85,671 Philippine farmers in two weeks (Sep 2025). In India, about ₹948 crore of crop-insurance claims sat unpaid in mid-2025, and some farmers waited up to 14 months.

The cruel part: the payout arrives after the next sowing season, when it can no longer buy seed. Assessment still depends on manual field surveys, and in the Philippines only about 1 in 10 farm households is insured at all.

GroundTruth: a farmer sends a phone photo and a note in their own language. Four agents on Google Cloud check it against satellite and rainfall evidence and settle it under the scheme rules in seconds, with every number traceable. The AI explains; a deterministic engine proves.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Different: Governments already run the two halves separately: satellite yield estimates for whole areas (India's YES-TECH) and farmer crop photos (CROPIC). Neither settles an individual claim. GroundTruth reconciles the farmer's photo with satellite evidence into one cited, deterministic decision per claim, explained back to the farmer.

Solves it by: Taking the survey off the critical path for corroborated claims; surveys go only to contradicted claims or unconfirmed top-ups. Best fit: individual localized-calamity claims (hail, inundation, post-harvest), which under India's PMFBY must be reported within 72 hours and today wait for a field inspection.

USP: A dramatic photo cannot outrun the satellite: assessed loss is capped by what remote sensing supports. Fraud is flagged, not silently paid or silently rejected. Works in the farmer's language.

Who it serves: smallholder farmers first; insurers and government schemes (India's PMFBY, Philippines' PCIC, Nepal's subsidised crop insurance) as the paying customers, with fewer surveys and fewer disputes.

List of features offered by the solution

Multilingual multimodal intake Gemini reads the field photo + note in Hindi, Marathi, Filipino, English and more

Satellite + weather corroboration Vegetation-index drop, rainfall anomaly and standing-water days per district

Deterministic adjudicator Covered peril, 7-day intimation, 33% threshold, corroboration, payout formula

Fair top-up Pays the satellite-proven loss now and books a survey for any larger reported loss, so local damage is never silently cut

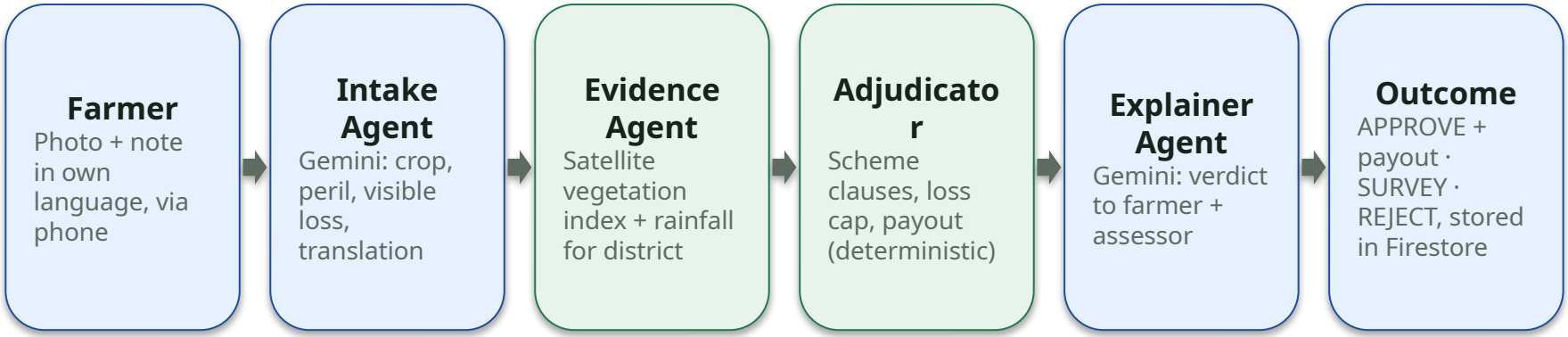
Fraud / mismatch flag Contradicted claims routed to physical survey

Explainable verdicts Farmer message in their language + assessor summary with cited clauses

Claims ledger Every claim, evidence pack and agent trace stored in Firestore

Never breaks Deterministic fallback if the model is unavailable

Process flow diagram or Use-case diagram

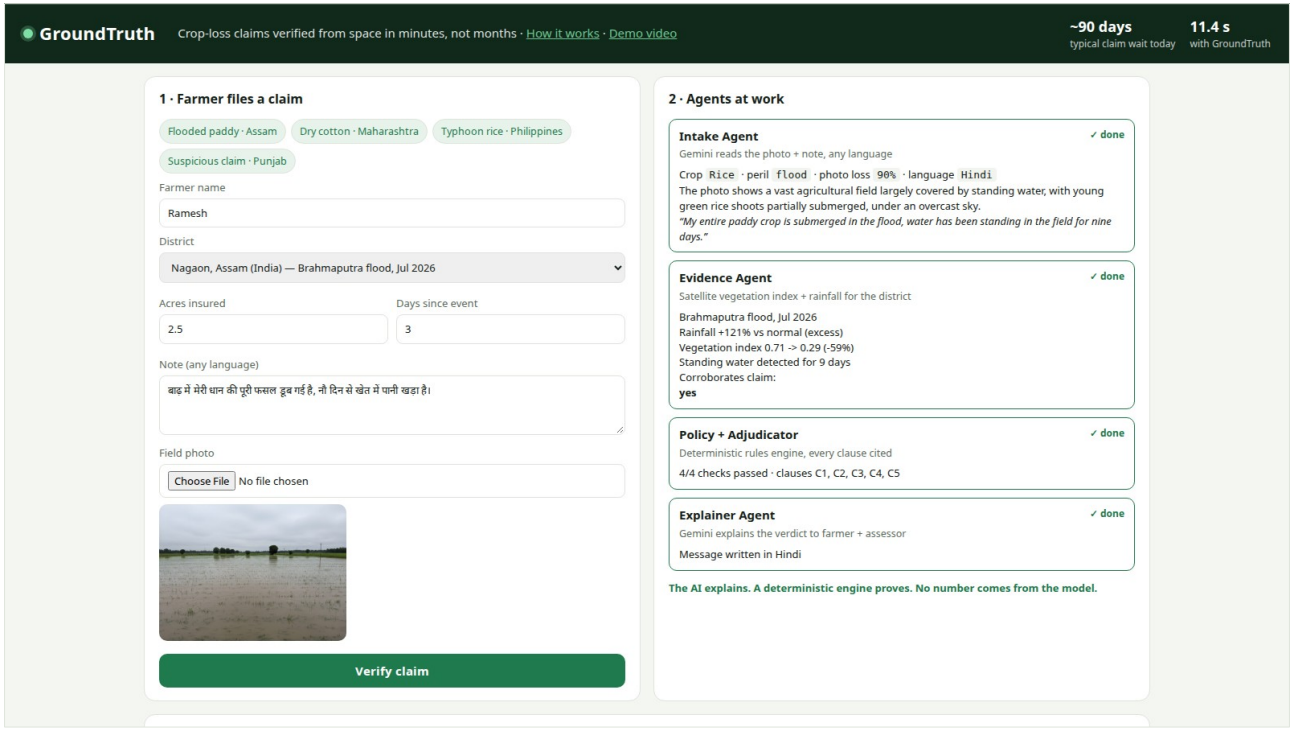


Blue = Gemini (reads, translates, explains)

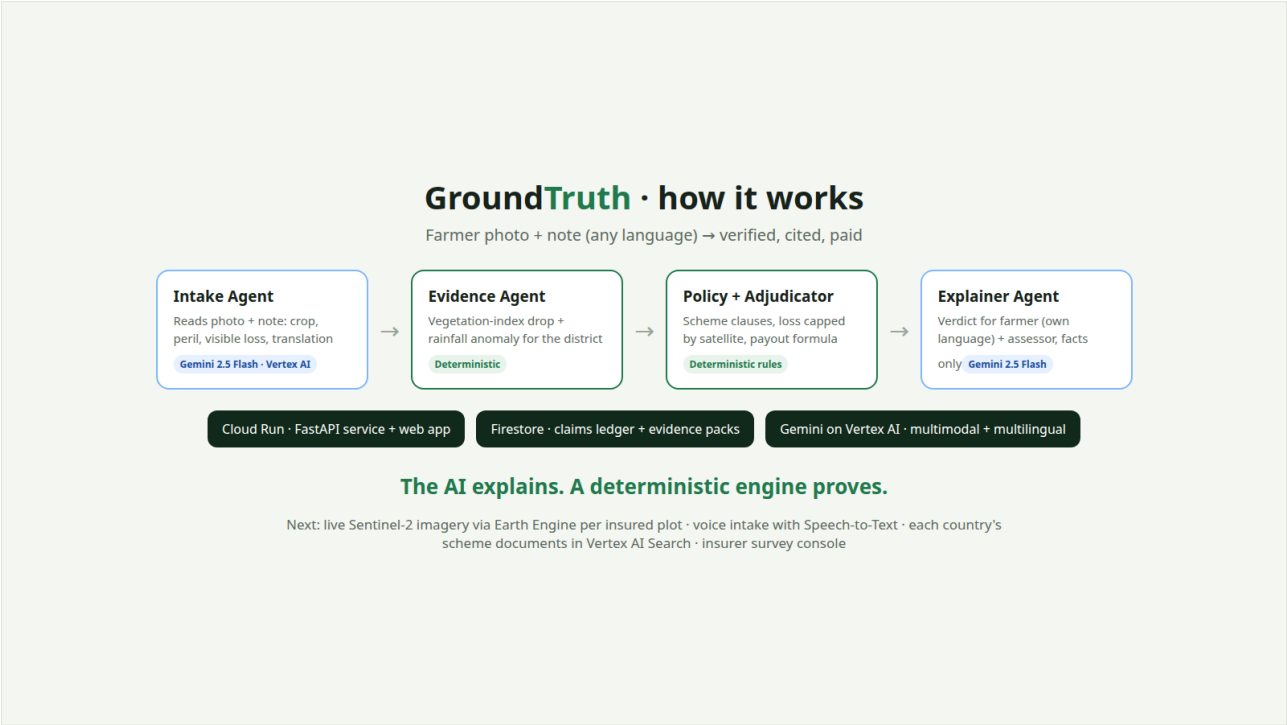
Green = deterministic engine (every number, every verdict)

Branch: photo/note contradicted by satellite + rainfall → fraud flag → physical survey queue

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies to be used in the solution

Gemini 2.5 Flash on Vertex AI multimodal intake agent and multilingual explainer agent

Cloud Run serverless FastAPI service + web app, deployed from source

Firestore (Firebase) claims ledger: claim, evidence pack, decision, agent trace

Cloud Build + Artifact Registry container build and image storage

Cloud Text-to-Speech narration for the demo

Python, FastAPI, vanilla JS no heavy front-end framework; accessible, mobile-friendly UI

Roadmap on GCP Earth Engine (Sentinel-2 per plot), Speech-to-Text voice intake, Vertex AI Search over scheme documents, BigQuery for portfolio analytics

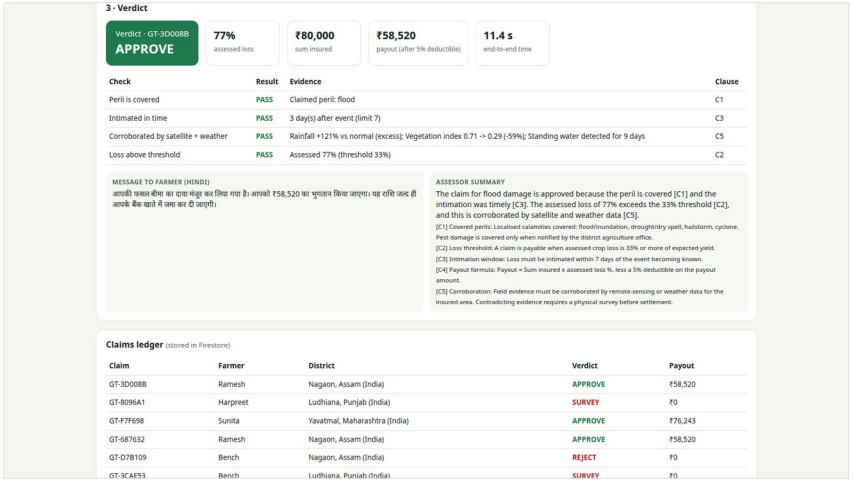
Estimated implementation cost (optional)

Per claim: 2 Gemini Flash calls (one with an image) + one Firestore write: well under one US cent per claim at list prices.

Platform: Cloud Run scales to zero; a pilot district of 10,000 claims a season runs for tens of US dollars of cloud cost.

Biggest saving: field surveys avoided for corroborated claims, which today cost insurers far more per claim than the cloud does.

Snapshots of the prototype



3 · Verdict

Verdict · GT-3D008B
APPROVE

77% assessed loss | ₹80,000 sum insured | ₹58,520 payout (after 5% deductible) | 11.4 s end-to-end time

Check	Result	Evidence	Clause
Peril is covered	PASS	Claimed peril: flood	C1
Intimated in time	PASS	3 day(s) after event (limit 7)	C3
Corroborated by satellite + weather	PASS	Rainfall +121% vs normal (excess); Vegetation index 0.71 -> 0.29 (-59%); Standing water detected for 9 days	C5
Loss above threshold	PASS	Assessed 77% (threshold 33%)	C2

MESSAGE TO FARMER (HINDI)

आपकी फसल बीमा का दावा मंज़ूर कर लिया गया है। आपको ₹58,520 का भुगतान किया जाएगा। यह राशि जल्द ही आपके बैंक खाते में जमा कर दी जाएगी।

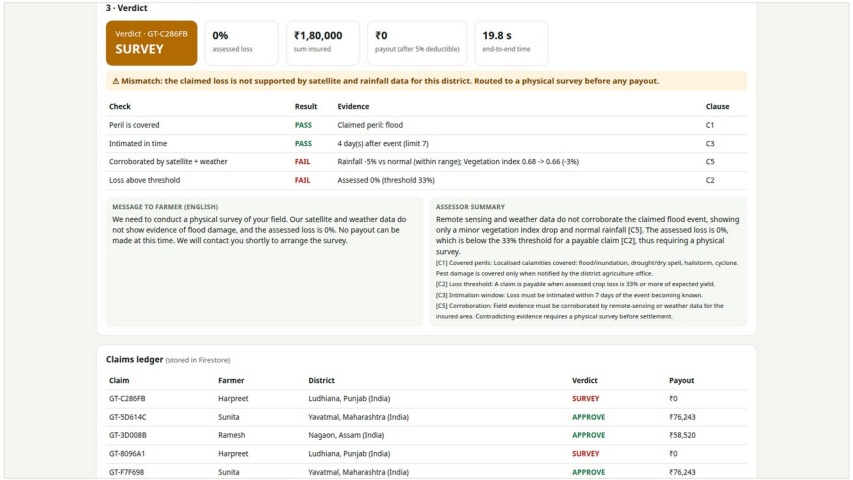
ASSESSOR SUMMARY

The claim for flood damage is approved because the peril is covered [C1] and the intimation was timely [C3]. The assessed loss of 77% exceeds the 33% threshold [C2], and this is corroborated by satellite and weather data [C5].

[C1] Covered perils: Localised calamities covered: flood/inundation, drought/dry spell, hailstorm, cyclone. Pest damage is covered only when notified by the district agriculture office.
[C2] Loss threshold: A claim is payable when assessed crop loss is 33% or more of expected yield.
[C3] Intimation window: Loss must be intimated within 7 days of the event becoming known.
[C4] Payout formula: Payout = Sum insured x assessed loss %, less a 5% deductible on the payout amount.
[C5] Corroboration: Field evidence must be corroborated by remote-sensing or weather data for the insured area. Contradicting evidence requires a physical survey before settlement.

Claims ledger (stored in Firestore)

Claim	Farmer	District	Verdict	Payout
GT-3D008B	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520
GT-8096A1	Harpreet	Ludhiana, Punjab (India)	SURVEY	₹0
GT-F7F698	Sunita	Yavatmal, Maharashtra (India)	APPROVE	₹76,243
GT-687632	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520
GT-D7B109	Bench	Nagaon, Assam (India)	REJECT	₹0
GT-3CAE53	Bench	Ludhiana, Punjab (India)	SURVEY	₹0



3 · Verdict

Verdict · GT-C286FB
SURVEY

0% assessed loss | ₹1,80,000 sum insured | ₹0 payout (after 5% deductible) | 19.8 s end-to-end time

⚠ Mismatch: the claimed loss is not supported by satellite and rainfall data for this district. Routed to a physical survey before any payout.

Check	Result	Evidence	Clause
Peril is covered	PASS	Claimed peril: flood	C1
Intimated in time	PASS	4 day(s) after event (limit 7)	C3
Corroborated by satellite + weather	FAIL	Rainfall -5% vs normal (within range); Vegetation index 0.68 -> 0.66 (-3%)	C5
Loss above threshold	FAIL	Assessed 0% (threshold 33%)	C2

MESSAGE TO FARMER (ENGLISH)

We need to conduct a physical survey of your field. Our satellite and weather data do not show evidence of flood damage, and the assessed loss is 0%. No payout can be made at this time. We will contact you shortly to arrange the survey.

ASSESSOR SUMMARY

Remote sensing and weather data do not corroborate the claimed flood event, showing only a minor vegetation index drop and normal rainfall [C5]. The assessed loss is 0%, which is below the 33% threshold for a payable claim [C2], thus requiring a physical survey.

[C1] Covered perils: Localised calamities covered: flood/inundation, drought/dry spell, hailstorm, cyclone. Pest damage is covered only when notified by the district agriculture office.
[C2] Loss threshold: A claim is payable when assessed crop loss is 33% or more of expected yield.
[C3] Intimation window: Loss must be intimated within 7 days of the event becoming known.
[C5] Corroboration: Field evidence must be corroborated by remote-sensing or weather data for the insured area. Contradicting evidence requires a physical survey before settlement.

Claims ledger (stored in Firestore)

Claim	Farmer	District	Verdict	Payout
GT-C286FB	Harpreet	Ludhiana, Punjab (India)	SURVEY	₹0
GT-5D614C	Sunita	Yavatmal, Maharashtra (India)	APPROVE	₹76,243
GT-3D008B	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520
GT-8096A1	Harpreet	Ludhiana, Punjab (India)	SURVEY	₹0
GT-F7F698	Sunita	Yavatmal, Maharashtra (India)	APPROVE	₹76,243

Prototype Performance report/Benchmarking

Scenario	Expected	Verdict	Assessed loss	Payout (₹)	Latency (s)	Language
Flood, Assam (Hindi)	APPROVE	APPROVE	77.0%	58,520	8.0	Hindi
Drought, Maharashtra (Marathi)	APPROVE	APPROVE	52.8%	76,243	11.6	Marathi
Typhoon, Philippines (Filipino)	APPROVE	APPROVE	67.9%	61,925	11.2	Tagalog
Unsupported claim, Punjab	SURVEY	SURVEY	0.0%	0	7.5	English
Late report, Assam (20 days)	REJECT	REJECT	77.0%	0	9.5	English

Result: 5/5 verdicts correct on live Cloud Run · median end-to-end 9.5s per claim (vs weeks to months today) · 6/6 engine unit tests pass · identical inputs give identical decisions.

Additional Details/Future Development (if any)

Live data: plot-level Sentinel-2 NDVI via Earth Engine, using field boundaries such as Google's AnthroKrishi maps; Sentinel-1 radar so monsoon cloud cover does not blind flood checks; gridded rainfall feeds.

Voice intake: Speech-to-Text so farmers can simply speak the claim.

Scheme knowledge: Vertex AI Search over each country's scheme documents, so policy clauses are retrieved and cited per programme.

Insurer console: survey queue for flagged claims, portfolio analytics in BigQuery.

Scale: tenant-per-insurer isolation, least-privilege IAM, regional deployment across JAPAC.

Note: satellite and rainfall values in this prototype are synthetic, clearly labelled sample signals; the scheme is a model area-yield scheme.

Sources: Kathmandu Post (2 Nov 2025; 7 Sep 2026); Malaya Business Insight / Philippine Dept. of Agriculture (2 Oct 2025); IASgyan summary of PMFBY data and IIM study (2025); PhilRice survey via SEARCA (2015).

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub repository: https://github.com/lokithelearner/groundtruth

Demo video: https://groundtruth-673378516921.asia-south1.run.app/static/GroundTruth_Demo.mp4

Live prototype: https://groundtruth-673378516921.asia-south1.run.app

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Thank you

